Table 2

Source-dependent forecasting, attack-family classification and time-to-attack performance

Dataset	Model	True-event windows	Forecast discrimination			Attack-family prediction		Conditional ETA	Joint fit
			H5 AP	H10 AP	H15 AP	Accuracy	Macro F1	MAE (min)	NLL
CIC-IDS2017	CR-LSTM	1,710	**0.7492**	**0.9034**	**0.9358**	**0.9105**	**0.7252**	2.470	**0.6117**
	CR-TCN	1,710	0.7256	0.8953	0.9341	0.8643	0.6915	**2.364**	0.6256
UNSW-NB15	CR-LSTM	1,665	0.7288	0.9017	**0.9396**	0.7045	**0.5309**	2.605	0.6879
	CR-TCN	1,665	**0.7332**	**0.9033**	0.9383	**0.7351**	0.5139	**2.602**	**0.6712**
Pooled	CR-LSTM	3,375	**0.7248**	**0.8991**	**0.9361**	**0.8089**	**0.7958**	2.537	0.6493
	CR-TCN	3,375	0.7174	0.8939	0.9318	0.8006	0.7864	**2.482**	**0.6481**

Notes: AP = average precision; MAE = mean absolute error; NLL = negative log-likelihood. Bold indicates the better of the two models within each dataset.

The event-window column counts true future-event windows; pooled horizon AP is evaluated over 18,839 aligned decision windows.

Future attack-family and ETA metrics are conditional on a true future event. Source-specific support differs and some attack families are absent in individual sources.

Synthetic NAF v2 development validation. Source-specific and pooled aggregate scores are reported as saved; no confidence intervals are implied.